F statistics

2024-11-11

```
res_dir <- "../results/s06_b_f_stats/"
```

Load

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

This script uses admixtools (ref Maier) to calculate F3 and F4 statistics, using the full global dataset, comprising of 2744 individuals and 4 loci.

Remove all populations with only one individual

```
pop_tbl <- table(mad_bor_g1000_sum$population)
to_remove <- which(pop_tbl == 1)
to_remove_pops <- names(to_remove)
mad_bor_g1000_sum <-
  mad_bor_g1000_sum %>% filter(!population %in% to_remove_pops)
```

Update labels

```
# # rename mad_bor_g1000_sum$population "Sakalava Tsimihety" to "Sakalava/Tsimihety"
# mad_bor_g1000_sum <- mad_bor_g1000_sum %>% mutate(population = case_when(
#   population == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
#   TRUE ~ population
# ))
# # rename mad_bor_g1000_sum$population "Diego" to "Diego Mixed"
# mad_bor_g1000_sum <- mad_bor_g1000_sum %>% mutate(population = case_when(
#   population == "Diego" ~ "Diego Mixed",
#   TRUE ~ population
# ))
```

Split Merina

```
# # Load fastmixture results
qmats <-
  readRDS("~/Dropbox/gitdata/malagasy_diversity/data/intermediate/fastmixture/fastmixture_m.RData")
# tidy with the gt
mad_bor_g1000_sum <- augment(get_q_matrix(qmats, k = 2, run = 1),
  data = mad_bor_g1000_sum
)
# filter to merina
merina <- mad_bor_g1000_sum %>% filter(population == "Merina")
# split groups
# .Q3 representing Asian ancestry and .Q2 representing African ancestry
merina$population <- ifelse(merina$.Q1 > 0.5, "MerinaA", "MerinaB")
merina <- merina[, c("id", "population")]
#merina$id <- sub("RR.*", "", merina$id)

# match V2 and merina$id and replace the population column in merina_fam with
# the population column in merina
matching_m <- match(mad_bor_g1000_sum$id, merina$id)
mad_bor_g1000_sum$population <- ifelse(!is.na(matching_m),
  merina$population[matching_m], mad_bor_g1000_sum$population
)

# count number of MerinaA
cat("Merina group > 0.5", length(which(mad_bor_g1000_sum$population == "MerinaA")))
```

```
## Merina group > 0.5 23
```

```
cat("Merina group < 0.5", length(which(mad_bor_g1000_sum$population == "MerinaB")))
```

```
## Merina group < 0.5 11
```

Group by population

```r
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

Create f2 blocks directory

```r
if (dir.exists("../data/intermediate/f2")) {
  f2_dir <- "../data/intermediate/f2"
} else {
  dir.create("../data/intermediate/f2")
  gt_extract_f2(mad_bor_g1000_sum, outdir = "../data/intermediate/f2")
  f2_dir <- "../data/intermediate/f2"
}
```

Read in precomputed f2's

```r
f2_blocks <- f2_from_precomp(f2_dir) # warning herre consider afprod
```

```
## i Reading precomputed data for 137 populations...
## i Reading f2 data for pair 1 out of 9453...i Reading f2 data for pair 2 out of 9453...i Reading f2 da
```

```
## Warning in f2_from_precomp(f2_dir): Some f2-statistic estimates are negative
## across blocks. This is probably caused by too many missing or rare SNPs in
## populations with low sample size, which makes the f2 bias correction
## unreliable. Consider running 'f2_from_precomp' with 'afprod = TRUE'.
```

```r
dim(f2_blocks)
```

```
## [1]   137   137 45175
```

```r
count_snps(f2_blocks)
```

```
## [1] 197972
```

Test for admixture: F3(Malagasy; Asian, African)

```r
gc()
```

```
##             used   (Mb) gc trigger    (Mb)   max used    (Mb)
## Ncells   2715007  145.0    5820134   310.9    5820134   310.9
## Vcells 854831812 6521.9 2454554690 18726.8 2454554061 18726.8
```

```r
all_pops <- unique(mad_bor_g1000_sum$population)

malagasy <- c(
  "Betsimisaraka", "MerinaA", "MerinaB","Diego", "Sakalava_Tsimihety",
  "Betsileo", "Temoro", "Vezo", "Mikea"
)
asian <- c("Borneo", "Sumba", "Timor", "KHV", "CHS")
# removing ,"BEB","PJL"
# african <- c("YRI", "LWK", "GWD", "ESN", "MSL")
# Yoruba, Luhya, Gambian, Esan, Mende

african <- setdiff(all_pops, c(malagasy, asian))

# Running test directly crashes R session, instead we can run in a loop
#test <- qp3pop(f2_blocks, malagasy, asian, african) # A, B ; C

results <- vector("list", length(african))
names(results) <- african

for (c_pop in african) {
  message("Running C = ", c_pop)
  results[[c_pop]] <- qp3pop(
    f2_blocks,
    malagasy,
    asian,
    c_pop
  )
  gc()
}
```

```
## Running C = PJL

## Running C = GWD

## Running C = ESN

## Running C = BEB

## Running C = MSL

## Running C = YRI

## Running C = LWK

## Running C = Botswana_Kalanga

## Running C = Zimbabwe_Remba

## Running C = CAR_Gbaya

## Running C = CAR_DzangaShangaPeople
```

```
## Running C = CAR_Mpiemo

## Running C = CAR_Banda

## Running C = CAR_Nzakara

## Running C = CAR_Zande

## Running C = SouthAfrica_ColouredWellington

## Running C = SouthAfrica_Swazi

## Running C = SouthAfrica_Karretjie

## Running C = SouthAfrica_ColouredColesberg

## Running C = SouthAfrica_Xhosa

## Running C = Namibia_Damara

## Running C = Namibia_Himba

## Running C = Angola_Khwe

## Running C = Botswana_Kgalagadi

## Running C = SouthAfrica_Khomani

## Running C = SouthAfrica_SothoSouthern

## Running C = SouthAfrica_Zulu

## Running C = SouthAfrica_Bhaca

## Running C = SouthAfrica_Hlubi

## Running C = SouthAfrica_Mputhi

## Running C = Mozambique_Nhungwe

## Running C = Mozambique_Tsonga

## Running C = Mozambique_Ronga

## Running C = Mozambique_TongaMoz

## Running C = Mozambique_Chuwabu
```

```
## Running C = Mozambique_Tswa

## Running C = Mozambique_Chopi

## Running C = SouthAfrica_Ndebele

## Running C = Angola_Xun

## Running C = Namibia_Wambo

## Running C = SouthAfrica_SothoNorthern

## Running C = Botswana_TswanaBot

## Running C = Botswana_GuiGhanaKgal

## Running C = Lesotho_SothoSouthern

## Running C = Swaziland_Swazi

## Running C = DRC_Manyanga

## Running C = DRC_Yombe

## Running C = DRC_Ndibu

## Running C = DRC_MongoNkundo

## Running C = DRC_LubaLulua

## Running C = DRC_Ntandu

## Running C = DRC_Shi

## Running C = DRC_Tetela

## Running C = DRC_Yans

## Running C = DRC_Mbala

## Running C = DRC_Nandi

## Running C = DRC_Yaka

## Running C = DRC_Ngombe

## Running C = DRC_Pende
```

```
## Running C = DRC_Songe

## Running C = DRC_Kuba

## Running C = DRC_Lunda

## Running C = SouthAfrica_Tsonga

## Running C = SouthAfrica_Tswana

## Running C = SouthAfrica_Venda

## Running C = Rwanda_Nkore

## Running C = Uganda_Konzo

## Running C = Uganda_Kiga

## Running C = Uganda_Fumbira

## Running C = Uganda_Nkore

## Running C = Zambia_Chewa

## Running C = Zambia_Kaonde

## Running C = Zambia_Lozi

## Running C = Zambia_ShonaZam

## Running C = Zambia_Bemba

## Running C = Zambia_Tumbuka

## Running C = Zambia_TongaZam

## Running C = Tanzania_TanzaniaMixed

## Running C = Zanzibar_Swahili

## Running C = Zambia_Ngoni

## Running C = Zambia_Nsenga

## Running C = SouthAfrica_Coloured

## Running C = Angola_Vasekele
```

```
## Running C = Namibia_TsumkweKung

## Running C = Namibia_Herero

## Running C = Namibia_Nama

## Running C = Angola_Sekele

## Running C = Namibia_Kgalagadi

## Running C = DRC_Luozi

## Running C = Zambia_Tswana

## Running C = Zambia_Kalanga

## Running C = Zambia_Kgalagadi

## Running C = Zambia_Mbukushu

## Running C = Zambia_Mbunda

## Running C = Zambia_Kwangwa

## Running C = Zambia_Luyana

## Running C = Zambia_Nkoya

## Running C = Zambia_Nyengo

## Running C = Zambia_Shanjo

## Running C = Zambia_Kwamashi

## Running C = Zambia_Fwe

## Running C = DRC_Ding

## Running C = DRC_Wonk

## Running C = DRC_Lwer

## Running C = DRC_Nsong

## Running C = DRC_Mbuun

## Running C = DRC_Nzadi
```

```
## Running C = DRC_Hungan

## Running C = DRC_Lele

## Running C = DRC_Mpiin

## Running C = DRC_Ngwi

## Running C = DRC_LubaKasai

## Running C = DRC_Ngong

## Running C = DRC_Sakata

## Running C = DRC_Mpe

## Running C = DRC_NunuBobangi

## Running C = DRC_Mbudza

## Running C = DRC_LubaKatanga

## Running C = DRC_Rega

## Running C = DRC_BanguBangu

## Running C = SouthAfrica_Pedi

## Running C = SouthAfrica_Sotho

## Running C = Botswana_Ghanzi
```

```r
test <- do.call(rbind, results)

colnames(test) <- c("Malagasy", "Asian", "African", "est", "se", "z", "p")

test$Asian <- sub("BOR", "Borneo", test$Asian)
test$Asian <- sub("CHS", "Han, China", test$Asian)
test$Asian <- sub("KHV", "Kinh, Vietnam", test$Asian)
test$Asian <- sub("PJL", "Punjabi, Pakistan", test$Asian)
test$Asian <- sub("BEB", "Bengali, Bengal", test$Asian)

test$African <- sub("ESN", "Esan, Nigeria", test$African)
test$African <- sub("YRI", "Yoruba, Nigeria", test$African)
test$African <- sub("GWD", "Mandinka, The Gambia", test$African)
test$African <- sub("LWK", "Luhye, Kenya", test$African)
test$African <- sub("MSL", "Mende, Sierra Leone", test$African)

test$Malagasy <- sapply(
  strsplit(test$Malagasy, "_"),
  function(x) paste(x, collapse = " ")
)

print(test)
```

```
## # A tibble: 5,535 x 7
##    Malagasy      Asian         African     est        se     z     p
##    <chr>         <chr>         <chr>     <dbl>     <dbl> <dbl> <dbl>
##  1 Betsileo      Borneo        PJL     0.00751 0.0000668  112.     0
##  2 Betsileo      Han, China    PJL     0.00813 0.0000702  116.     0
##  3 Betsileo      Kinh, Vietnam PJL     0.00801 0.0000682  117.     0
##  4 Betsileo      Sumba         PJL     0.00877 0.0000708  124.     0
##  5 Betsileo      Timor         PJL     0.00967 0.0000867  112.     0
##  6 Betsimisaraka Borneo        PJL     0.0138  0.0000885  155.     0
##  7 Betsimisaraka Han, China    PJL     0.0142  0.0000907  157.     0
##  8 Betsimisaraka Kinh, Vietnam PJL     0.0142  0.0000893  159.     0
##  9 Betsimisaraka Sumba         PJL     0.0145  0.0000900  161.     0
## 10 Betsimisaraka Timor         PJL     0.0150  0.000103   146.     0
## # i 5,525 more rows
```

Admixture f3 statistics were used in the configuration f3(Malagasy; Asian, African) to examine which comparative African and Asian populations show the greatest similarity to the 'source' populations of the Malagasy.

Examine min F3 values for each Malagasy popualtion

```
obj <- test %>%
  group_by(Malagasy) %>%
  slice(which.min(est))
obj
```

```
## # A tibble: 9 x 7
## # Groups:   Malagasy [9]
##   Malagasy           Asian  African              est        se      z     p
##   <chr>              <chr>  <chr>              <dbl>     <dbl>  <dbl> <dbl>
## 1 Betsileo           Borneo Mozambique_Ronga -0.0124  0.0000925 -134.      0
## 2 Betsimisaraka      Borneo Mozambique_Chopi -0.0120  0.0000810 -149.      0
## 3 Diego              Borneo Mozambique_Tswa  -0.0113  0.0000828 -137.      0
## 4 MerinaA            Borneo Mozambique_Tswa  -0.00786 0.0000780 -101.      0
## 5 MerinaB            Borneo Mozambique_Ronga -0.0116  0.000113  -102.      0
## 6 Mikea              Borneo Mozambique_Ronga -0.0105  0.000111   -94.7     0
## 7 Sakalava Tsimihety Borneo Mozambique_Chopi -0.0122  0.0000811 -150.      0
## 8 Temoro             Borneo Mozambique_Ronga -0.0120  0.000104  -115.      0
## 9 Vezo               Borneo Mozambique_Chopi -0.0120  0.0000829 -145.      0
```

```
write.table(
  file = paste0(res_dir, "minimum_f3_malagasy_pops"),
  obj, row.names = FALSE, quote = FALSE
)
```

```
# Examine z-scores remembering that z-score is affected by population size,
# so interpretation should be careful
min_z <- test %>%
  group_by(Malagasy) %>%
  slice(which.min(z))
min_z
```

```
## # A tibble: 9 x 7
## # Groups:   Malagasy [9]
##   Malagasy           Asian  African       est        se     z     p
##   <chr>              <chr>  <chr>       <dbl>     <dbl> <dbl> <dbl>
## 1 Betsileo           Borneo DRC_Shi  -0.0115  0.0000551 -209.     0
## 2 Betsimisaraka      Borneo DRC_Shi  -0.0110  0.0000516 -213.     0
## 3 Diego              Borneo DRC_Shi  -0.0102  0.0000485 -211.     0
## 4 MerinaA            Borneo DRC_Ngwi -0.00753 0.0000641 -118.     0
## 5 MerinaB            Borneo DRC_Ngwi -0.0109  0.0000676 -162.     0
## 6 Mikea              Borneo DRC_Ngwi -0.00991 0.0000614 -161.     0
## 7 Sakalava Tsimihety Borneo DRC_Shi  -0.0112  0.0000508 -220.     0
## 8 Temoro             Borneo DRC_Shi  -0.0110  0.0000537 -205.     0
## 9 Vezo               Borneo DRC_Shi  -0.0110  0.0000546 -201.     0
```

```
mad_bor_g1000_sum %>% filter(population == "DRC_Ngwi") %>% nrow()
```

```
## [1] 67
```

```
mad_bor_g1000_sum %>% filter(population == "DRC_Shi") %>% nrow()
```

```
## [1] 129
```

```
mad_bor_g1000_sum %>% filter(population == "Mozambique_Ronga") %>% nrow()
```

```
## [1] 2
```

```
mad_bor_g1000_sum %>% filter(population == "Mozambique_Chopi") %>% nrow()
```

```
## [1] 4
```

```
mad_bor_g1000_sum %>% filter(population == "Mozambique_Tswa") %>% nrow()
```

```
## [1] 4
```

```
write.table(
  file = paste0(res_dir, "minimum_z-score_malagasy_pops"),
  min_z, row.names = FALSE, quote = FALSE
)
```

F3 values grouped descending all combinations

```
# update labels
# rename test$Malagasy "Sakalava Tsimihety" to "Sakalava/Tsimihety"
test <- test %>% mutate(Malagasy = case_when(
  Malagasy == "Sakalava Tsimihety" ~ "Sakalava/Tsimihety",
  TRUE ~ Malagasy
))
```

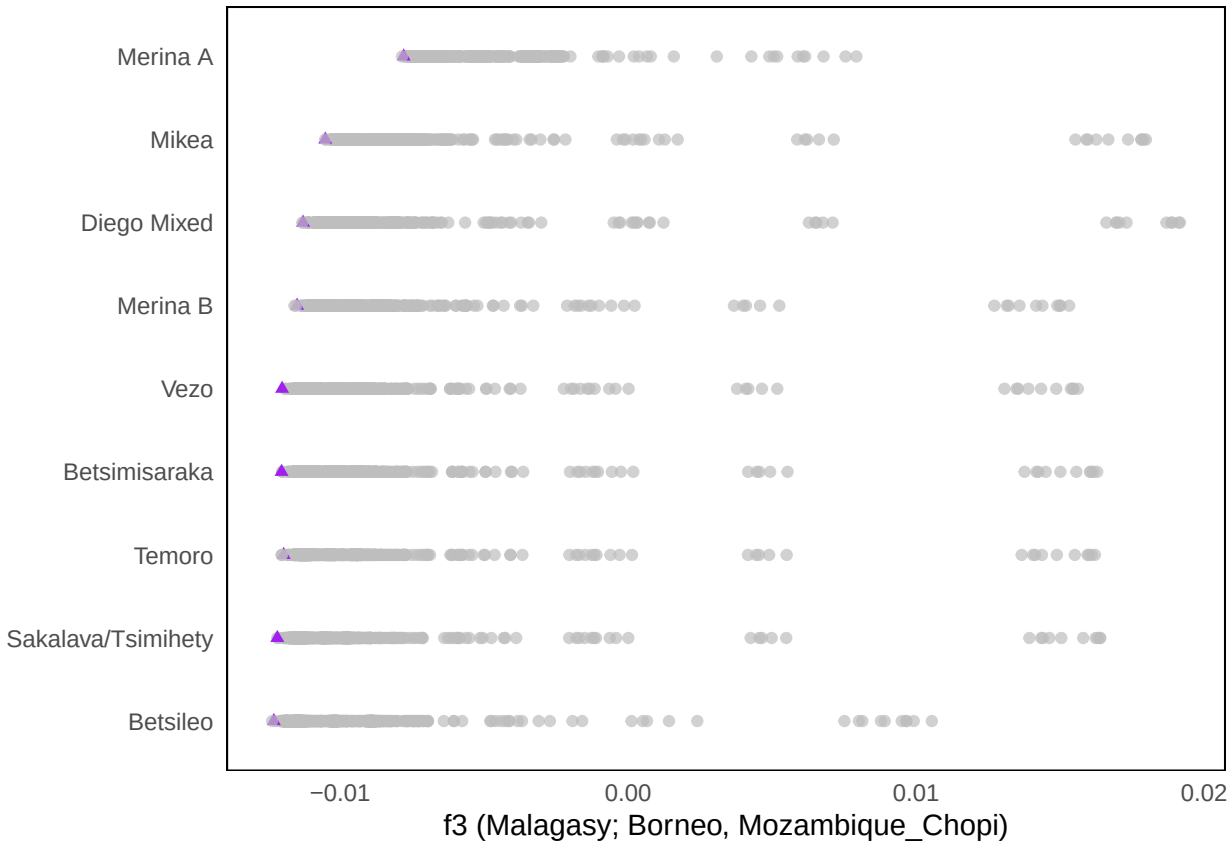
```r
# rename test$Malagasy "Diego" to "Diego Mixed"
test <- test %>% mutate(Malagasy = case_when(
  Malagasy == "Diego" ~ "Diego Mixed",
  TRUE ~ Malagasy
))
# rename test$Malagasy "MerinaA" to "Merina A"
test <- test %>% mutate(Malagasy = case_when(
  Malagasy == "MerinaA" ~ "Merina A",
  TRUE ~ Malagasy
))
# rename test$Malagasy "MerinaB" to "Merina B"
test <- test %>% mutate(Malagasy = case_when(
  Malagasy == "MerinaB" ~ "Merina B",
  TRUE ~ Malagasy
))


test_grouped <- test %>%
  select(est, se, z, Malagasy, Asian, African) %>%
  reframe(est = est, se = se, z, Malagasy, Asian, African)

test_grouped$highlight <- "NA"
# test_grouped$highlight[test_grouped$Asian == "Borneo"] <- "Borneo"
# test_grouped$highlight[test_grouped$African == "Mozambique_Chopi"] <- "Mozambique_Chopi"
# find which rows in test_grouped has $African == "Mozambique_Chopi" and $Asian == "Borneo"
test_grouped$highlight[which(test_grouped$African == "Mozambique_Chopi" & test_grouped$Asian == "Borneo
test_grouped$highlight <- as.factor(test_grouped$highlight)

test_grouped %>%
  group_by(Malagasy) %>%
  arrange(desc(est)) %>%
  ggplot(aes(
    x = est, y = reorder(Malagasy, est),
    color = highlight, shape = highlight, alpha = highlight
  )) +
  geom_point() +
  labs(
    x = "f3 (Malagasy; Borneo, Mozambique_Chopi)", y = NULL,
    color = element_blank(), shape = element_blank(),
    alpha = element_blank()
  ) +
  scale_shape_manual(
    values = c(
      "NA" = 19, # "Borneo" = 3, "Mozambique_Chopi" = 15,
      "Borneo and Mozambique_Chopi" = 17
    ),
    breaks = c(
      # "Borneo", "Mozambique_Chopi",
      "Borneo and Mozambique_Chopi"
    )
  ) +
  scale_color_manual(
    values = c(
```

```
    "NA" = "grey", # "Borneo" = "lightpink", "Mozambique_Chopi" = "lightblue",
    "Borneo and Mozambique_Chopi" = "purple"
  ),
  breaks = c(
    # "Borneo", "Mozambique_Chopi",
    "Borneo and Mozambique_Chopi"
  )
) +
scale_alpha_manual(
  values = c(
    "NA" = 0.7, # "Borneo" = 0.7, "Mozambique_Chopi" = 0.7,
    "Borneo and Mozambique_Chopi" = 1
  ),
  breaks = c(
    # "Borneo", "Mozambique_Chopi",
    "Borneo and Mozambique_Chopi"
  )
) +
theme_minimal() +
theme(
  #axis.text.x = element_blank(),
  legend.position = "none",
  axis.line = element_blank(),
  panel.grid.major = element_blank(),
  panel.grid.minor = element_blank(),
  panel.background = element_rect(fill = "white")
)
```

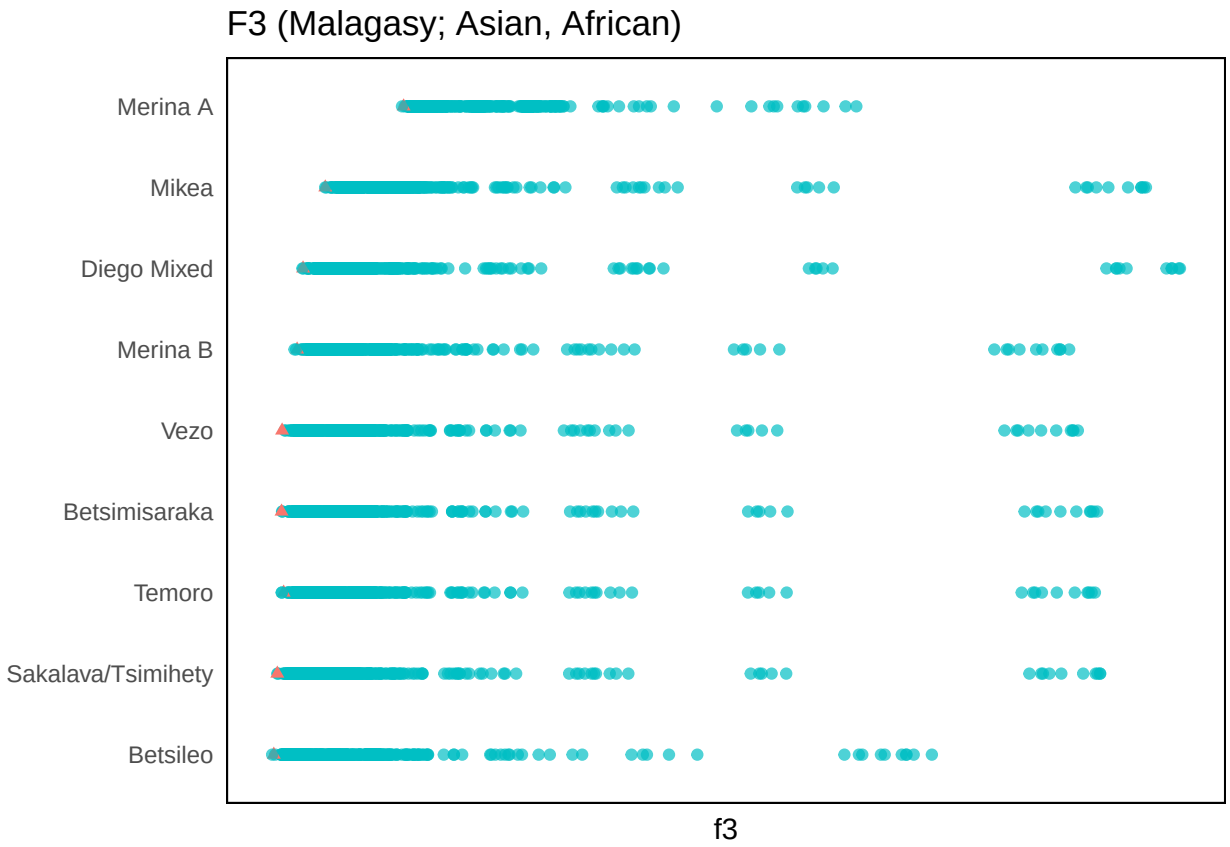



```
ggsave(
  filename = paste0(res_dir, "F3_all_combinations.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

```
test_grouped$highlight <- "NA"
# find which rows in test_grouped has $African == "Mozambique_Chopi" and $Asian == "Borneo"
test_grouped$highlight[which(test_grouped$African == "Mozambique_Chopi" & test_grouped$Asian == "Borneo
test_grouped$highlight <- as.factor(test_grouped$highlight)

test_grouped %>%
  group_by(Malagasy) %>%
  arrange(desc(est)) %>%
  ggplot(aes(
    x = est, y = reorder(Malagasy, est),
    color = highlight, shape = highlight,
    alpha = highlight
  )) +
  geom_point() +
  labs(
    title = "F3 (Malagasy; Asian, African)", x = "f3",
    y = NULL, color = element_blank(),
    shape = element_blank()
  ) +
  scale_shape_manual(
    values = c(
```

```
    "NA" = 19, # "Borneo" = 3, "Mozambique_Chopi" = 15,
    "Borneo and Mozambique_Chopi" = 17
  ),
  breaks = c(
    # "Borneo", "Mozambique_Chopi",
    "Borneo and Mozambique_Chopi"
  )
) +
guides(color = "none", shape = "none", alpha = "none") +
scale_alpha_manual(
  values = c(
    "NA" = 0.7, # "Borneo" = 0.7, "Mozambique_Chopi" = 0.7,
    "Borneo and Mozambique_Chopi" = 1
  ),
  breaks = c(
    # "Borneo", "Mozambique_Chopi",
    "Borneo and Mozambique_Chopi"
  )
) +
theme_minimal() +
theme(
  axis.text.x = element_blank(),
  legend.position = "right",
  axis.line = element_blank(),
  panel.grid.major = element_blank(),
  panel.grid.minor = element_blank(),
  panel.background = element_rect(fill = "white")
)
```



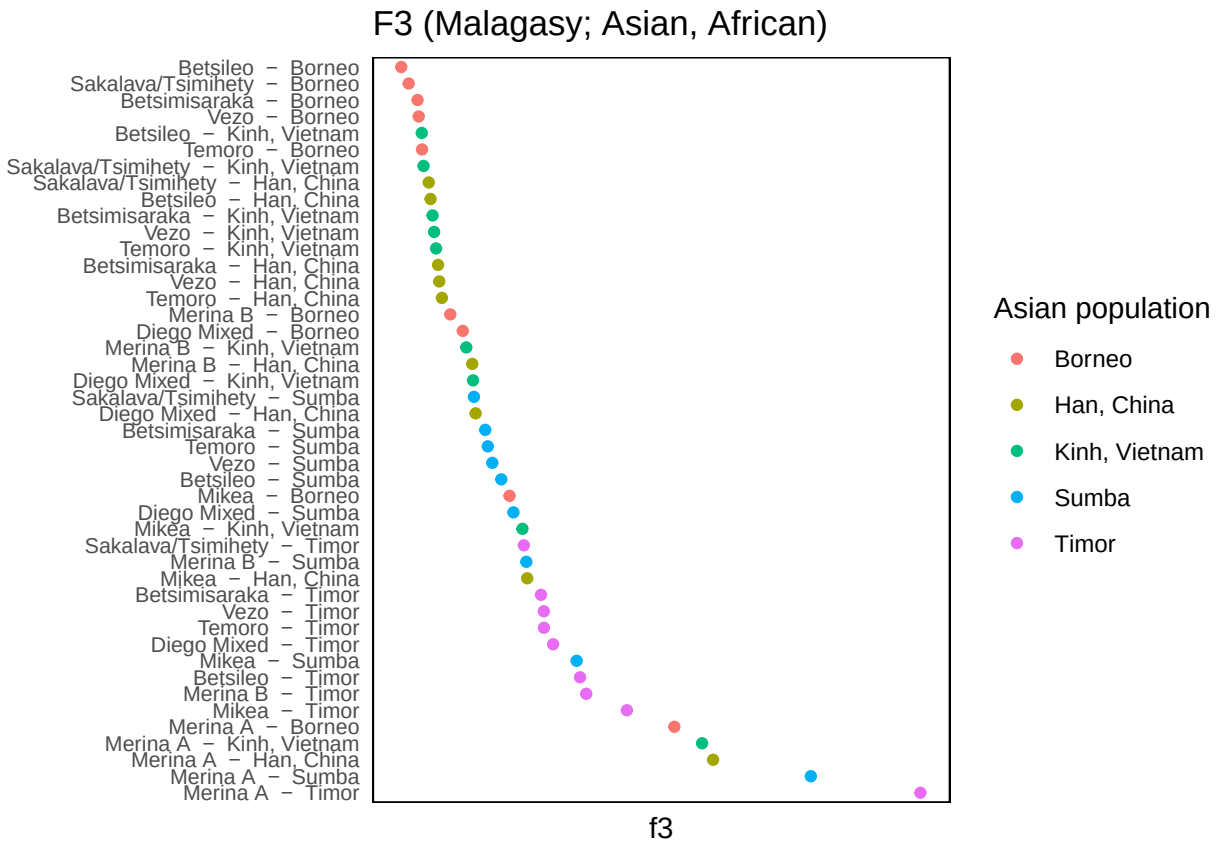
F3 values descending (with Yoruba)

```
Y <- test %>%
  filter(African == "Mozambique_Chopi") %>%
  select(est, se, z, Malagasy, Asian)

Y <- Y[order(Y$est), ]
Y$F3 <- paste(Y$Malagasy, " - ", Y$Asian)
Y$F3 <- factor(Y$F3, levels = Y$F3[order(Y$est, decreasing = TRUE)])

ggplot(Y, aes(
  x = est, y = F3, colour = Asian,
  xmax = est - (se / 2), xmin = est + (se / 2)
)) +
  geom_point() +
  # geom_errorbar()+
  labs(
    title = "F3 (Malagasy; Asian, African)",
    x = "f3", y = NULL, colour = "Asian population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
```

```
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```

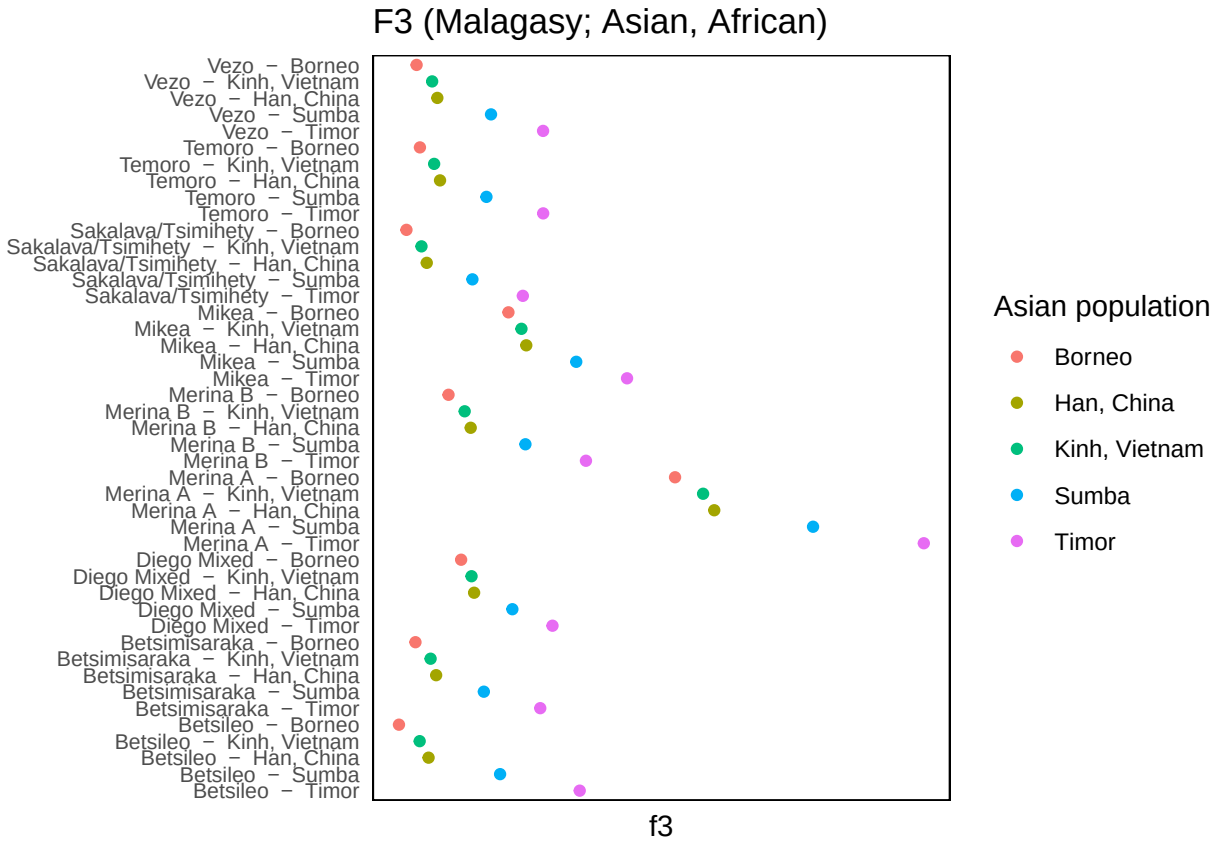


F3 values for each Malagasy ordered (with Yoruba)

```
Y <- test %>%
  filter(African == "Mozambique_Chopi") %>%
  select(est, se, Malagasy, Asian) %>%
  reframe(est = est, se = se, Malagasy, Asian)

Y$F3 <- paste(Y$Malagasy, " - ", Y$Asian)
Y <- Y %>%
  arrange(desc(est)) %>%
  select(est, se, Malagasy, Asian, F3) %>%
  group_by(Malagasy) %>%
  mutate(F3 = forcats::fct_inorder(F3))
```

```
ggplot(Y, aes(x = est, y = F3, colour = Asian)) +
  geom_point() +
  labs(
    title = "F3 (Malagasy; Asian, African)", x = "f3",
    y = NULL, colour = "Asian population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```

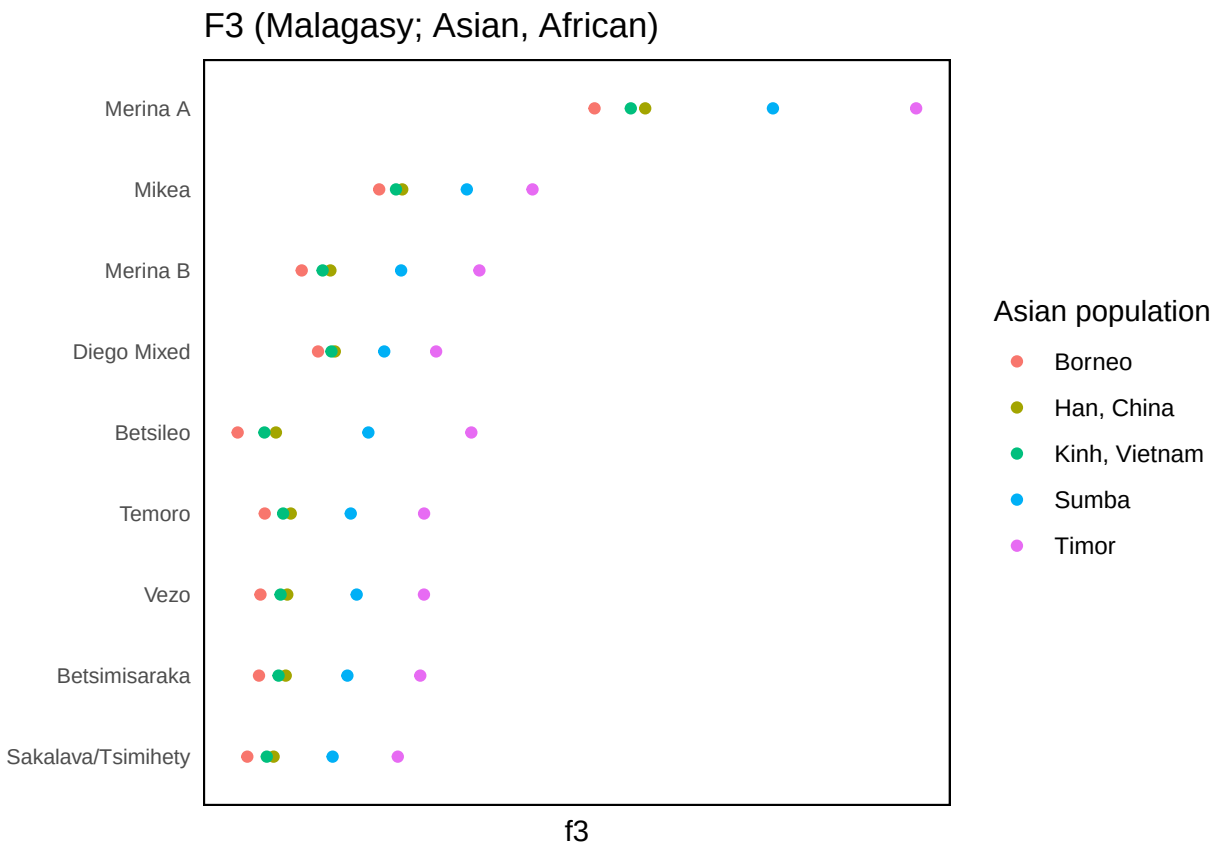



```
ggsave(
  filename = paste0(res_dir, "F3_Asian_source.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

F3 values for each Malagasy grouped (with Yoruba)

```
Y <- test %>%
  filter(African == "Mozambique_Chopi") %>%
  select(est, se, z, Malagasy, Asian) %>%
  reframe(est = est, se = se, z, Malagasy, Asian)


Y %>%
  group_by(Malagasy) %>%
  arrange(desc(est)) %>%
  ggplot(aes(x = est, y = reorder(Malagasy, est), colour = Asian)) +
  geom_point() +
  labs(
    title = "F3 (Malagasy; Asian, African)", x = "f3", y = NULL,
    colour = "Asian population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```



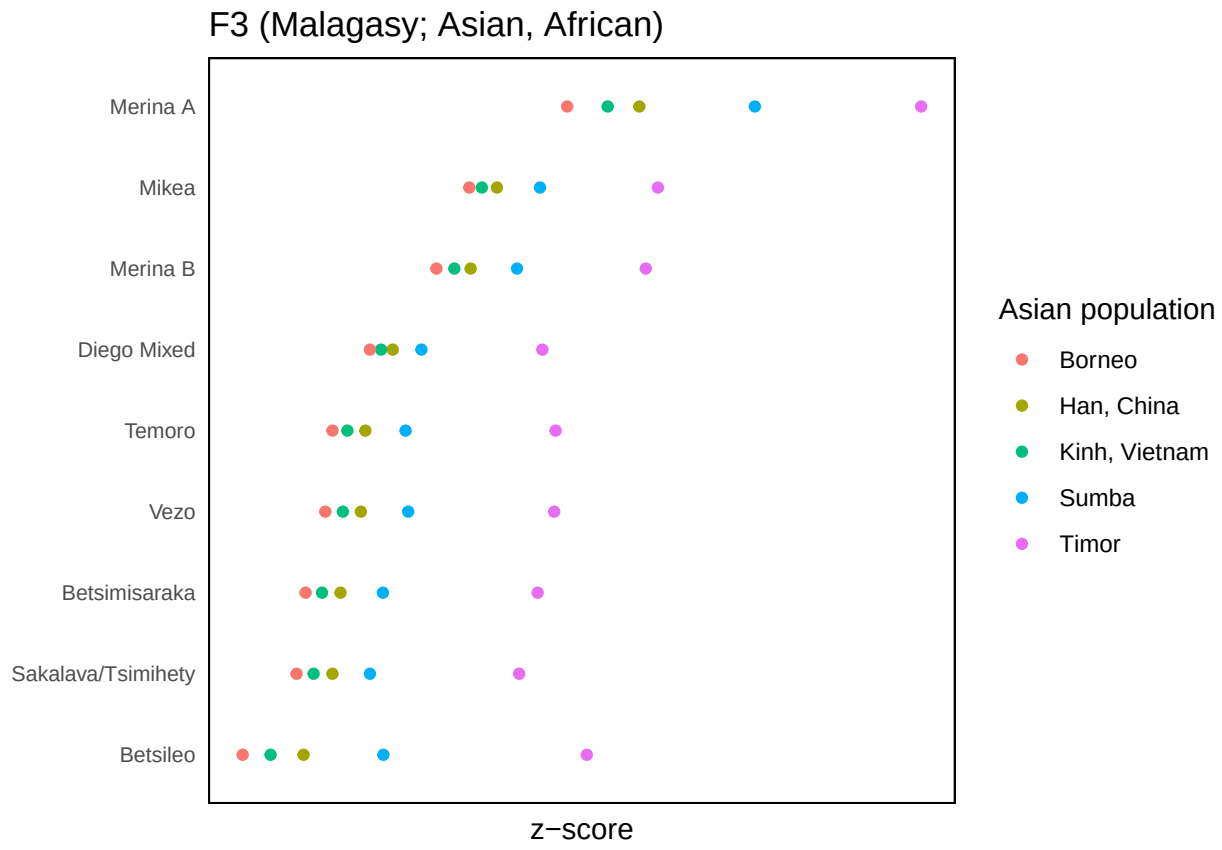

```
ggsave(
  filename = paste0(res_dir, "F3_Asian_source_per_malagasy_pop.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

Also check z-scores

```
Y <- test %>%
  filter(African == "Mozambique_Chopi") %>%
  select(est, se, z, Malagasy, Asian) %>%
  reframe(est = est, se = se, z, Malagasy, Asian)


Y %>%
  group_by(Malagasy) %>%
  arrange(desc(z)) %>%
  ggplot(aes(x = z, y = reorder(Malagasy, z), colour = Asian)) +
  geom_point() +
  labs(
    title = "F3 (Malagasy; Asian, African)", x = "z-score", y = NULL,
    colour = "Asian population"
  ) +
  theme_minimal() +
```

```
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```




```
ggsave(
  filename = paste0(res_dir, "F3_Asian_source_per_malagasy_pop_z.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

F3 values descending (with Borneo)

```
# San, Yoruba, Luhya, Gambian, Esan, Mende
test$African <- sub("ESN", "Esan, Nigeria", test$African)
test$African <- sub("KHM_SA", "Khomani San, South Africa", test$African)
test$African <- sub("YRI", "Yoruba, Nigeria", test$African)
test$African <- sub("GWD", "Mandinka, The Gambia", test$African)
```

```r
test$African <- sub("LWK", "Luhye, Kenya", test$African)
test$African <- sub("MSL", "Mende, Sierra Leone", test$African)

Y <- test %>%
  filter(Asian == "Borneo") %>%
  select(est, se, z, Malagasy, African)

Y <- Y[order(Y$est), ]
Y$F3 <- paste(Y$Malagasy, " - ", Y$African)
Y$F3 <- factor(Y$F3, levels = Y$F3[order(Y$est, decreasing = TRUE)])

plt <- ggplot(Y, aes(
  x = est, y = F3,
  colour = African, xmax = est - (se / 2), xmin = est + (se / 2)
)) +
  geom_point() +
  # geom_errorbar()+
  labs(
    title = "F3 (Malagasy; Asian, African)", x = "f3", y = NULL,
    colour = "African population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```

F3 values for each Malagasy ordered (with Borneo)

```r
Y <- test %>%
  filter(Asian == "Borneo") %>%
  select(est, se, Malagasy, African) %>%
  reframe(est = est, se = se, Malagasy, African)

Y$F3 <- paste(Y$Malagasy, " - ", Y$African)
Y <- Y %>%
  arrange(desc(est)) %>%
  select(est, se, Malagasy, African, F3) %>%
  group_by(Malagasy) %>%
  mutate(F3 = forcats::fct_inorder(F3))

plt <- ggplot(Y, aes(x = est, y = F3, colour = African)) +
  geom_point() +
  labs(
    title = "F3 (Malagasy; Asian, African)", x = "f3", y = NULL,
    colour = "African population"
```

```
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )

# ggsave(
#  filename = paste0(res_dir, "F3_African_source.jpg"),
#  width = 7, height = 5, units = "in", dpi = 300
# )
```

#F3 values for each Malagasy grouped (with Borneo)

```
Y <- test %>%
  filter(Asian == "Borneo") %>%
  select(est, se, z, Malagasy, African) %>%
  reframe(est = est, se = se, z, Malagasy, African)


plt <- Y %>%
  group_by(Malagasy) %>%
  arrange(desc(est)) %>%
  ggplot(aes(x = est, y = reorder(Malagasy, est), colour = African)) +
  geom_point() +
  labs(
    title = "F3 (Malagasy; Asian, African)", x = "f3", y = NULL,
    colour = "African population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )

# ggsave(
#  filename = paste0(res_dir, "F3_African_source_per_malagasy_pop.jpg"),
#  width = 7, height = 5, units = "in", dpi = 300
# )
```

Also check z-scores

```
Y <- test %>%
  filter(Asian == "Borneo") %>%
  select(est, se, z, Malagasy, African) %>%
  reframe(est = est, se = se, z, Malagasy, African)


plt <- Y %>%
  group_by(Malagasy) %>%
  arrange(desc(z)) %>%
  ggplot(aes(x = z, y = reorder(Malagasy, z), colour = African)) +
  geom_point() +
  labs(
    title = "F3 (Malagasy; Asian, African)", x = "z-score", y = NULL,
    colour = "African population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )

# ggsave(
#  filename = paste0(res_dir, "F3_African_source_per_malagasy_pop_z.jpg"),
#  width = 7, height = 5, units = "in", dpi = 300
# )
```

Outgroup f3's

[@brucato2016a] used an outgroup f3 statistic to investigate the closest Asian population to the Malagasy. An f3 to assess the closest outgroup from a list of alternatives for an admixed population is not accurate, because there is not an equal branch length to the common ancestor. However, here we we will use F3(Asian, Malagasy; African) as in [@brucato2016a] to replicate their result and check our data.

```
african <- c("YRI")

test2 <- qp3pop(f2_blocks, african, asian, malagasy)
colnames(test2) <- c("African", "Asian", "Malagasy", "est", "se", "z", "p")


test2$Asian <- sub("BOR", "Borneo", test2$Asian)
test2$Asian <- sub("CHS", "Han Chinese", test2$Asian)
test2$Asian <- sub("KHV", "Kinh Vietnamese", test2$Asian)
test2$Asian <- sub("PJL", "Punjabi", test2$Asian)
test2$Asian <- sub("BEB", "Bengali", test2$Asian)
```

```r
test2$Malagasy <- sapply(
  strsplit(test2$Malagasy, "_"),
  function(x) paste(x, collapse = " ")
)

print(test2)
```

```
## # A tibble: 45 x 7
##    African Asian       Malagasy              est        se     z     p
##    <chr>   <chr>       <chr>               <dbl>     <dbl> <dbl> <dbl>
##  1 YRI     Borneo      Betsileo           0.0296 0.000115   257.     0
##  2 YRI     Borneo      Betsimisaraka      0.0210 0.0000895  235.     0
##  3 YRI     Borneo      Diego              0.0173 0.0000772  223.     0
##  4 YRI     Borneo      MerinaA            0.0406 0.000154   263.     0
##  5 YRI     Borneo      MerinaB            0.0234 0.000109   214.     0
##  6 YRI     Borneo      Mikea              0.0206 0.0000971  212.     0
##  7 YRI     Borneo      Sakalava Tsimihety 0.0204 0.0000864  236.     0
##  8 YRI     Borneo      Temoro             0.0211 0.0000920  229.     0
##  9 YRI     Borneo      Vezo               0.0220 0.0000959  229.     0
## 10 YRI     Han Chinese Betsileo           0.0291 0.000115   252.     0
## # i 35 more rows
```

```r
test2 %>%
  group_by(Malagasy) %>%
  slice(which.max(est))
```

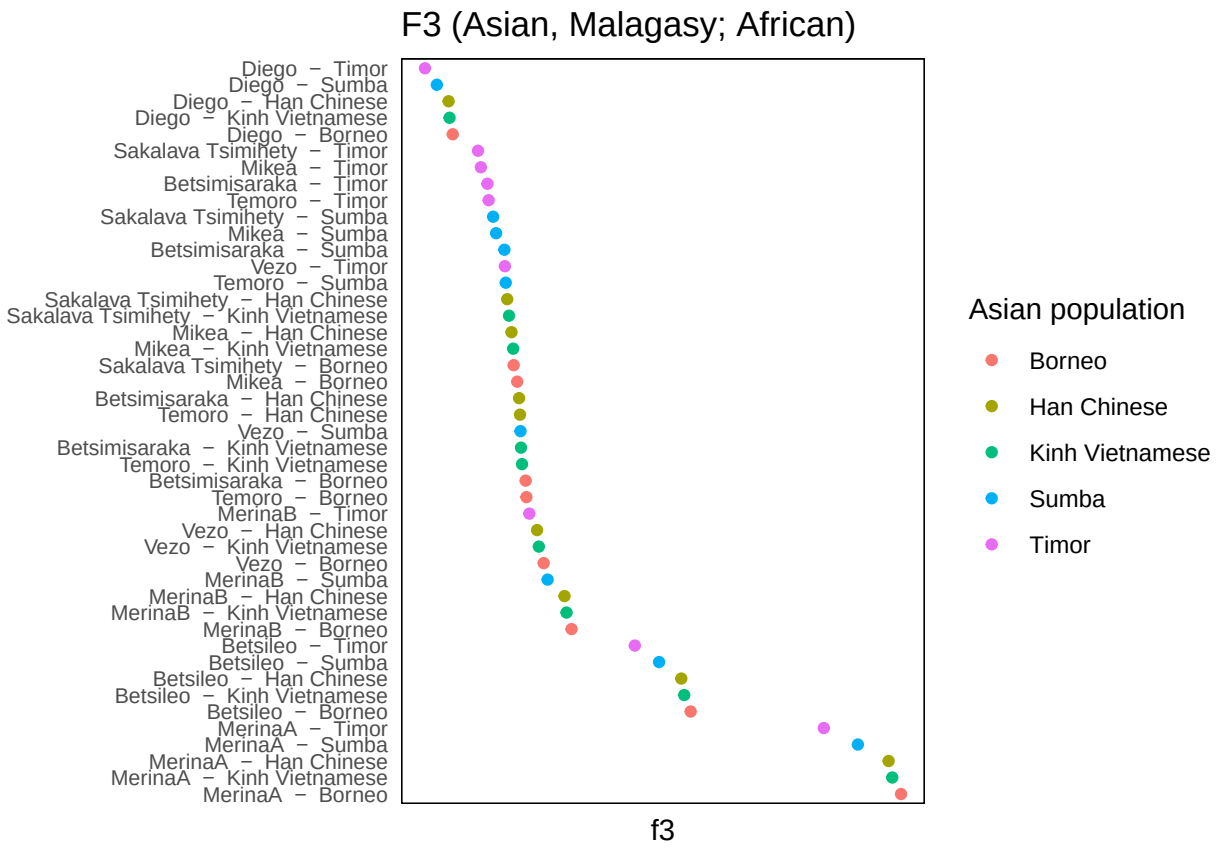
```
## # A tibble: 9 x 7
## # Groups:   Malagasy [9]
##   African Asian  Malagasy              est        se     z     p
##   <chr>   <chr>  <chr>               <dbl>     <dbl> <dbl> <dbl>
## 1 YRI     Borneo Betsileo           0.0296 0.000115   257.     0
## 2 YRI     Borneo Betsimisaraka      0.0210 0.0000895  235.     0
## 3 YRI     Borneo Diego              0.0173 0.0000772  223.     0
## 4 YRI     Borneo MerinaA            0.0406 0.000154   263.     0
## 5 YRI     Borneo MerinaB            0.0234 0.000109   214.     0
## 6 YRI     Borneo Mikea              0.0206 0.0000971  212.     0
## 7 YRI     Borneo Sakalava Tsimihety 0.0204 0.0000864  236.     0
## 8 YRI     Borneo Temoro             0.0211 0.0000920  229.     0
## 9 YRI     Borneo Vezo               0.0220 0.0000959  229.     0
```

```r
Y <- test2 %>%
  filter(African == "YRI") %>%
  select(est, se, Malagasy, Asian) %>%
  reframe(est = est, se = se, Malagasy, Asian)

Y <- Y[order(Y$est), ]
Y$F3 <- paste(Y$Malagasy, " - ", Y$Asian)
Y$F3 <- factor(Y$F3, levels = Y$F3[order(Y$est, decreasing = TRUE)])

ggplot(Y, aes(x = est, y = F3, colour = Asian)) +
  geom_point() + # , xmax = est - (se/2), xmin = est + (se/2)
  labs(
```

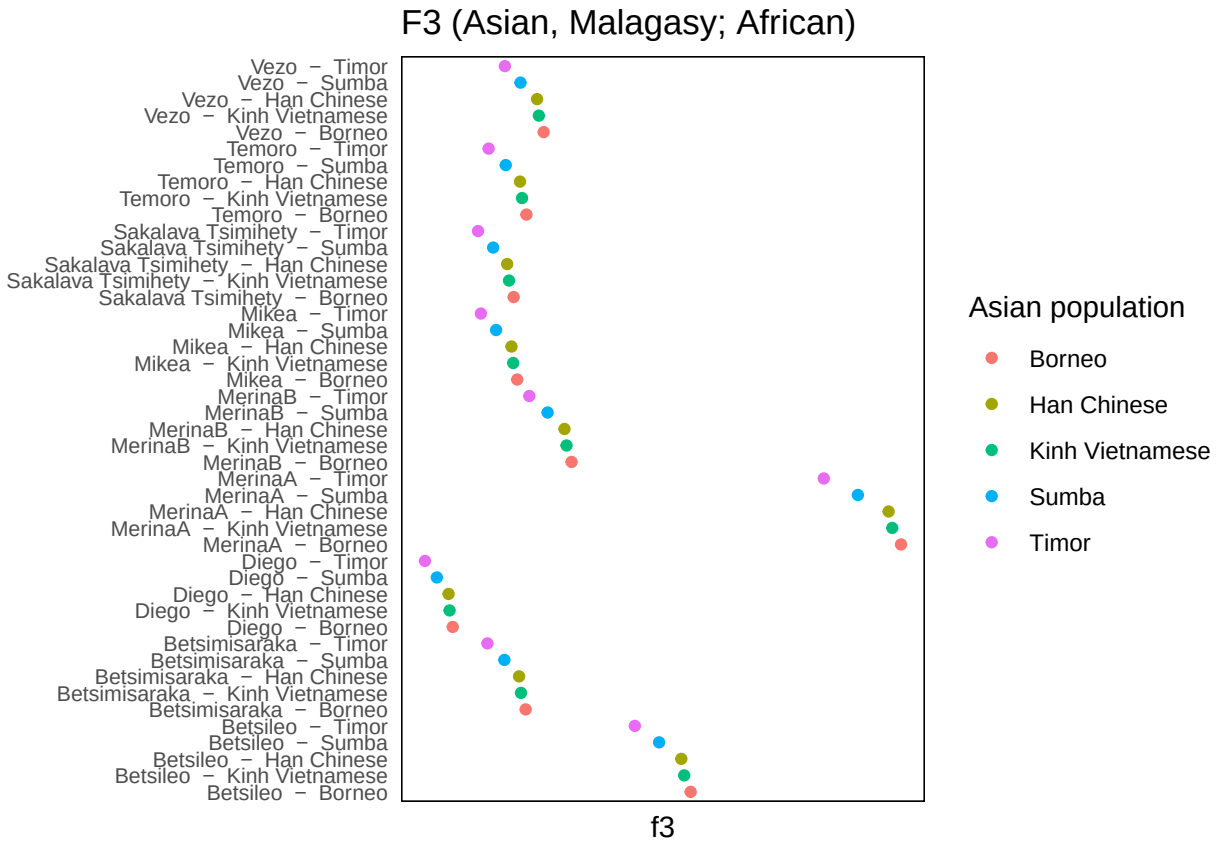
```
    title = "F3 (Asian, Malagasy; African)", x = "f3", y = NULL,
    colour = "Asian population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```




```
Y <- test2 %>%
  filter(African == "YRI") %>%
  select(est, se, Malagasy, Asian) %>%
  reframe(est = est, se = se, Malagasy, Asian)

Y$F3 <- paste(Y$Malagasy, " - ", Y$Asian)
Y <- Y %>%
  arrange(desc(est)) %>%
  select(est, se, Malagasy, Asian, F3) %>%
  group_by(Malagasy) %>%
  mutate(F3 = forcats::fct_inorder(F3))
```

```
ggplot(Y, aes(x = est, y = F3, colour = Asian)) +
  geom_point() +
  labs(
    title = "F3 (Asian, Malagasy; African)", x = "f3", y = NULL,
    colour = "Asian population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```



F4 ratio

To estimate the proportion of each ancestry source within the Malagasy population, we will use an f4 ratio in the following configurations:

f4(GBR,Gorilla;Malagasy,Borneo)/f4(GBR,Chimpanzee;Esan,Borneo)

f4(GBR,Gorilla;Malagasy,Esan)/f4(GBR,Chimpanzee;Borneo,Esan)

```r
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% ungroup()
chimp <- gt_load("../data/intermediate/pan_tro_liftover/chimp_ref_v2.gt")
f4_merged_data <- rbind(mad_bor_g1000_sum, chimp,
  as_is = FALSE, flip_strand = TRUE,
  quiet = FALSE, use_position = FALSE,
  backingfile = tempfile()
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 198196 reduced to 77993
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 77993 reduced to 77993
## ( 0 were flipped to match the reference set)
## ( 0 are ambiguous, of which 0 were removed)

##
## gen_tibble saved to /tmp/RtmpXCR00C/file37acf7a2b24bd.gt

## using FBM RDS: /tmp/RtmpXCR00C/file37acf7a2b24bd.rds

## with FBM backing file: /tmp/RtmpXCR00C/file37acf7a2b24bd.bk

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/tmp/RtmpXCR00C/file37acf7a2b24bd.gt')
```

```r
g1000 <- gt_load("../data/raw/data_input/1000_Genomes_b37_clean.gt")
# append population info
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")

pop_table_sub <- subset(pop_table, pop_table$IID %in% g1000$id)

matching_m <- match(g1000$id, pop_table_sub$IID)
g1000$population <- ifelse(!is.na(matching_m),
  pop_table_sub$V2[matching_m],
  g1000$population
)

GBR <- g1000 %>% filter(population == "GBR")

show_loci(GBR)$name <- paste(show_loci(GBR)$chromosome, show_loci(GBR)$position, sep = "_")
```

```
f4_merged_data <- rbind(f4_merged_data, GBR,
  as_is = FALSE, flip_strand = TRUE,
  quiet = FALSE, use_position = FALSE,
  backingfile = tempfile()
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 77993 reduced to 77993
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 670552 reduced to 77993
## ( 1777 were flipped to match the reference set)
## ( 3 are ambiguous, of which 3 were removed)

##
## gen_tibble saved to /tmp/RtmpXCR00C/file37acf1fc8c2df.gt

## using FBM RDS: /tmp/RtmpXCR00C/file37acf1fc8c2df.rds

## with FBM backing file: /tmp/RtmpXCR00C/file37acf1fc8c2df.bk

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/tmp/RtmpXCR00C/file37acf1fc8c2df.gt')
```

```
nrow(f4_merged_data)
```

```
## [1] 2792
```

```
f4_merged_data <- f4_merged_data %>% group_by(population)
```

Create f2 blocks directory

```
if (dir.exists("../data/intermediate/f2_outgroup")) {
  f2_dir <- "../data/intermediate/f2_outgroup"
} else {
  dir.create("../data/intermediate/f2_outgroup")
  gt_extract_f2(f4_merged_data,
    outdir = "../data/intermediate/f2_outgroup"
  )
}
```

Read in precomputed f2's

```
f2_dir <- "../data/intermediate/f2_outgroup"
f2_blocks <- f2_from_precomp(f2_dir, afprod = TRUE)
```

```
## i Reading precomputed data for 139 populations...
## i Reading ap data for pair 1 out of 9730...i Reading ap data for pair 2 out of 9730...i Reading ap da
```

```
dim(f2_blocks)
```

```
## [1]   139   139 24966
```

```
count_snps(f2_blocks)
```

```
## [1] 77911
```

```
gc()
```

```
##             used   (Mb) gc trigger    (Mb)   max used    (Mb)
## Ncells   4018748  214.7    7634202   407.8    5820134   310.9
## Vcells 500500203 3818.6 1963643752 14981.5 2454554061 18726.8
```

F4 ratio: estimating the proportion of African admixture

Sister taxon and outgroup on the left, target in the centre, and sister population and source of admixture on the right.

```
malagasy <- c(
  "Betsimisaraka", "MerinaA","MerinaB","Diego", "Sakalava_Tsimihety",
  "Betsileo", "Temoro", "Vezo", "Mikea"
)

pop_matrix <-
  as.matrix(data.frame("GBR", "chimp_ref", malagasy, "Borneo", "ESN"))

qp4_african_p <- qpf4ratio(f2_blocks, pop_matrix)

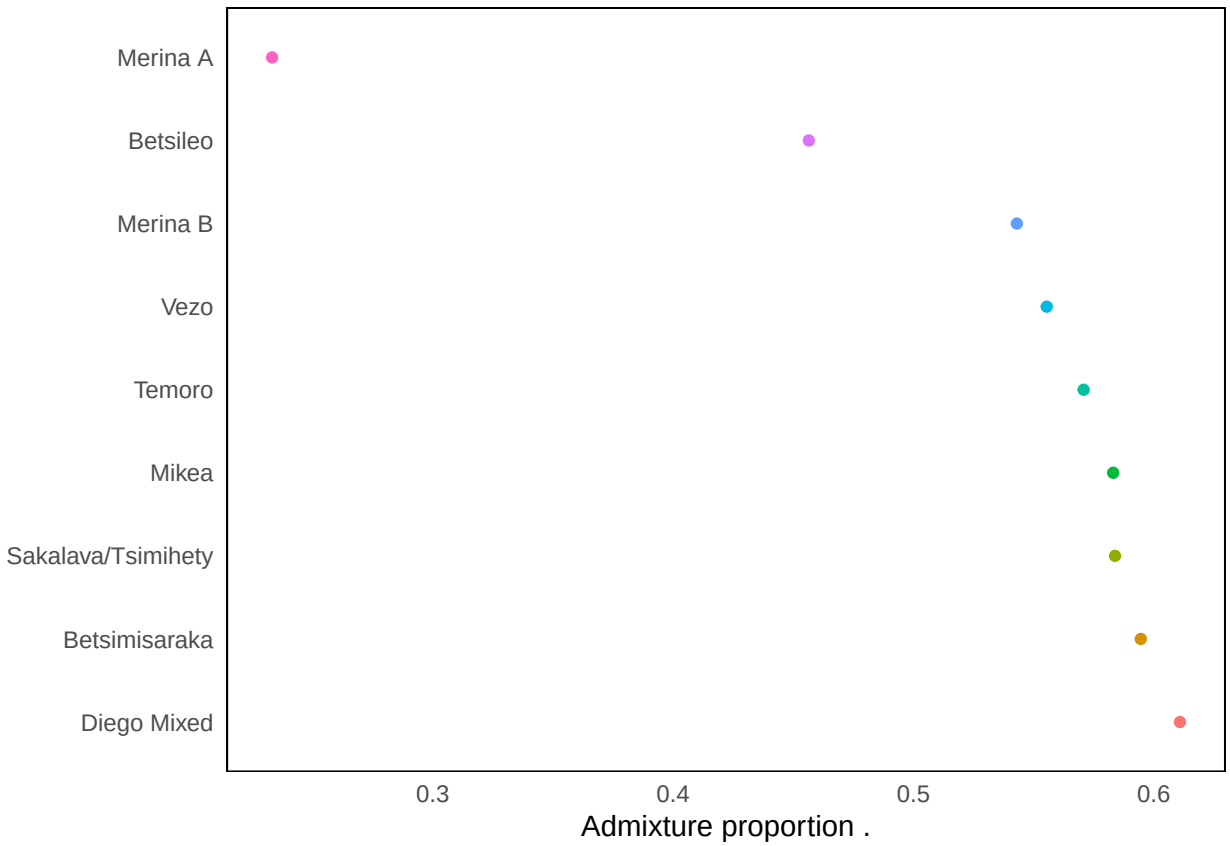
qp4_african_p$pop3 <- factor(qp4_african_p$pop3,
  levels = c(
    "Betsimisaraka", "MerinaA","MerinaB", "Diego",
    "Sakalava_Tsimihety", "Betsileo",
    "Temoro", "Vezo", "Mikea"
  )
)
# order by qp4_african_p$pop3 levels
qp4_african_p <- qp4_african_p %>% mutate(pop3 = forcats::fct_inorder(pop3))
# round alpha to 1 decimal place
qp4_african_p
```

```
## # A tibble: 9 x 8
##   pop1  pop2      pop3               pop4   pop5  alpha      se     z
##   <chr> <chr>     <fct>              <chr>  <chr> <dbl>   <dbl> <dbl>
## 1 GBR   chimp_ref Betsimisaraka      Borneo ESN   0.595 0.00903  65.9
## 2 GBR   chimp_ref MerinaA            Borneo ESN   0.233 0.0122   19.0
## 3 GBR   chimp_ref MerinaB            Borneo ESN   0.543 0.0129   42.2
## 4 GBR   chimp_ref Diego              Borneo ESN   0.611 0.00906  67.4
## 5 GBR   chimp_ref Sakalava_Tsimihety Borneo ESN   0.584 0.00893  65.4
## 6 GBR   chimp_ref Betsileo           Borneo ESN   0.456 0.00955  47.8
## 7 GBR   chimp_ref Temoro             Borneo ESN   0.571 0.00990  57.7
## 8 GBR   chimp_ref Vezo               Borneo ESN   0.555 0.0102   54.7
## 9 GBR   chimp_ref Mikea              Borneo ESN   0.583 0.0116   50.4
```

```r
# update labels
# rename qp4_african_p$pop3 "Sakalava Tsimihety" to "Sakalava/Tsimihety"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "Diego" to "Diego Mixed"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "Diego" ~ "Diego Mixed",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "MerinaA" to "Merina A"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "MerinaA" ~ "Merina A",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "MerinaB" to "Merina B"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "MerinaB" ~ "Merina B",
  TRUE ~ pop3
))

# FIX LEGEND
qp4_african_p <- qp4_african_p %>%
  arrange(desc(alpha)) %>%
  mutate(pop3 = forcats::fct_inorder(pop3))
ggplot(qp4_african_p, aes(x = alpha, y = pop3, colour = pop3, fill = pop3)) +
  geom_point() +
  labs(
    x = paste("Admixture proportion \u03b1"), y = NULL,
    colour = "Malagasy population", fill = "Malagasy population"
  ) +
  guides(colour = "none", fill = "none") +
  theme_minimal() +
  theme(
    # axis.text.x = element_blank(),
    #axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
```

```
    panel.background = element_rect(fill = "white")
  )
```




```
ggsave(
  filename = paste0(res_dir, "F4_African_source.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

Merina is lowest, as expected.

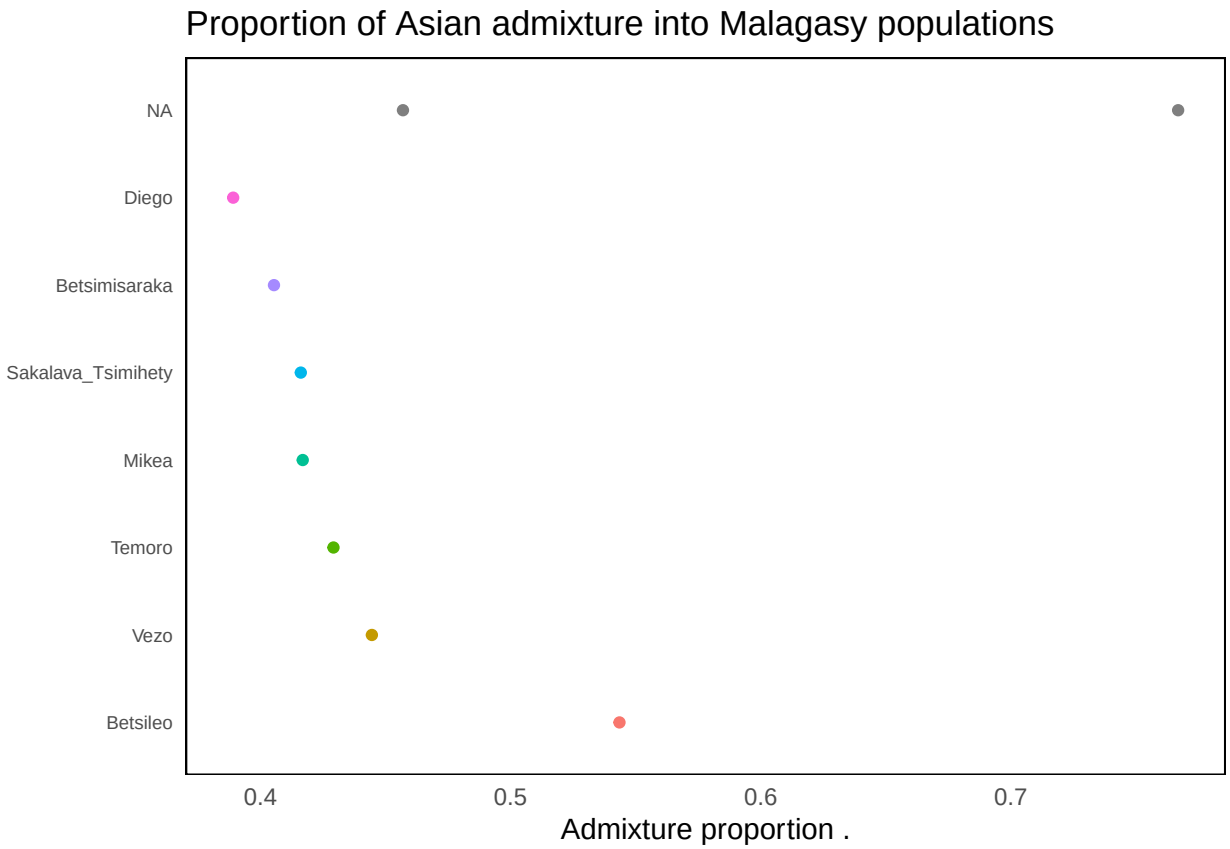
```
pop_matrix <-
  as.matrix(data.frame("GBR", "chimp_ref", malagasy, "ESN", "Borneo"))

qp4_asian_p <- qpf4ratio(f2_blocks, pop_matrix)
qp4_asian_p$pop3 <- factor(
  qp4_asian_p$pop3,
  c(
    "Betsimisaraka", "Merina", "Diego",
    "Sakalava_Tsimihety", "Betsileo", "Temoro",
    "Vezo", "Mikea"
  )
)
qp4_asian_p <- qp4_asian_p %>% mutate(pop3 = forcats::fct_inorder(pop3))
qp4_asian_p
```

```
## # A tibble: 9 x 8
##   pop1  pop2      pop3               pop4  pop5   alpha      se     z
##   <chr> <chr>     <fct>              <chr> <chr>  <dbl>   <dbl> <dbl>
## 1 GBR   chimp_ref Betsimisaraka      ESN   Borneo 0.405 0.00903  44.9
## 2 GBR   chimp_ref <NA>               ESN   Borneo 0.767 0.0122   62.6
## 3 GBR   chimp_ref <NA>               ESN   Borneo 0.457 0.0129   35.5
## 4 GBR   chimp_ref Diego              ESN   Borneo 0.389 0.00906  42.9
## 5 GBR   chimp_ref Sakalava_Tsimihety ESN   Borneo 0.416 0.00893  46.6
## 6 GBR   chimp_ref Betsileo           ESN   Borneo 0.544 0.00955  56.9
## 7 GBR   chimp_ref Temoro             ESN   Borneo 0.429 0.00990  43.3
## 8 GBR   chimp_ref Vezo               ESN   Borneo 0.445 0.0102   43.8
## 9 GBR   chimp_ref Mikea              ESN   Borneo 0.417 0.0116   36.0
```

Merina and Betsileo are highest, as expected.

```
qp4_asian_p <- qp4_asian_p %>%
  arrange(desc(alpha)) %>%
  mutate(pop3 = forcats::fct_inorder(pop3))
ggplot(qp4_asian_p, aes(x = alpha, y = pop3, colour = pop3, fill = pop3)) +
  geom_point() +
  labs(
    title = "Proportion of Asian admixture into Malagasy populations",
    x = paste("Admixture proportion \u03b1"), y = NULL,
    colour = "Malagasy population", fill = "Malagasy population"
  ) +
  guides(colour = "none", fill = "none") +
  theme_minimal() +
  theme(
    # axis.text.x = element_blank(),
    axis.text.y = element_text(size = 7),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```




```
ggsave(
  filename = paste0(res_dir, "F4_Asian_source.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

Recording R environment

Part of ensuring our workflow is reproducible is to keep track of the R environment. We will automatically save our R environment into a file called `r_environment.json`.